

The Role of Inventory Expectations in the Crude Oil Market: An Extension of Kilian (2009)

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1 Introduction

Understanding and evaluating the nature of oil price shocks and their effects on the responses of macroeconomic aggregates is a non-trivial objective. Specifically, two related issues arise. First, reverse causality between macroeconomic aggregates and oil prices raises concerns about the underlying cause-and-effect structure. Second, the price of oil is driven by distinct demand and supply shocks, each of which has different effects on both the real price of oil and the economy itself. Moreover, these shocks may affect economic activity not only directly, but also indirectly through the resulting changes in the real price of crude oil. Consequently, a naive approach in which oil price shocks are treated as exogenous in order to evaluate their effects on macroeconomic dynamics must be considered invalid, as the *ceteris paribus* condition is violated.

A seminal contribution to the resolution of the aforementioned concerns has been proposed by Kilian (2009). The author proposes a structural vector autoregressive (SVAR) model of the global crude oil market that explicitly disentangles the underlying sources of oil price fluctuations into three components: crude oil supply shocks, shocks to global demand for all industrial commodities, and demand shocks that are specific to the global crude oil market. Given this framework, the objectives are twofold: to identify the underlying demand and supply shocks in the global crude oil market and to evaluate their associated effects on the real price of oil itself, as well as to identify and estimate the effects of oil price shocks on macroeconomic aggregates, taking into account the aforementioned disentanglement.

Building on Kilian (2009), and after conducting a robustness check using an extended sample, this paper further develops the modeling of the global crude oil market. Specifically, it augments the baseline structural VAR by incorporating expected crude oil inventories. This extension addresses a central question: *does the explicit inclusion of inventory expectations improve the identification of oil price drivers by capturing part of the precautionary demand component?* The findings suggest that the extended model, in addition to uncovering an economically sound and theoretically consistent relationship between shocks to expected inventories and real oil prices, substantially improves shock identification. In particular, the variance of the cumulative effect of the residual shock on the real price of oil declines by 32%, and the gap between the cumulative responses implied by the two specifications is strongly correlated ($\rho = 0.80$) with the newly identified oil-specific precautionary demand shock. Crucially, this demonstrates that the extended specification successfully disentangles genuine precautionary demand from residual noise, explicitly capturing forward-looking dynamics that the baseline model conflates with unexplained shocks.

The paper is structured as follows. Section 2 introduces the SVAR model of Kilian (2009) and performs a robustness check by extending the estimation sample. Section 3 introduces and estimates the new SVAR model specification, relying on a newly constructed dataset of expected U.S. crude oil inventories to account for part of the precautionary-demand-driven oil price shock. Finally, Section 4 draws the conclusions.

2 Replication and Data Extension

In this section, the estimation proposed by Kilian (2009) is replicated and the sample period is extended, following a brief review of its theoretical foundations and methodology.

2.1 Theoretical Foundation and Methodology

The price of oil is determined in global markets, and the determinants of fluctuations in the real price of oil can be distinguished as follows: (a) shocks to the current physical availability of crude oil, capturing unexpected changes in global oil production; (b) shocks to the current demand for crude oil driven by fluctuations in the global business cycle, reflecting changes in overall economic

activity; and (c) shocks driven by shifts in precautionary demand for oil, which Kilian (2009) has interpreted as demand arising from uncertainty about shortfalls of expected oil supply relative to expected demand. Importantly, the latter plays a crucial role in capturing expectation-driven movements in the real price of oil that are not directly linked to contemporaneous changes in production or aggregate economic activity.

Given the aforementioned three-component theoretical decomposition, and considering the econometric issues related to treating oil price fluctuations as exogenous, Kilian (2009) introduced a methodology for decomposing unpredictable changes in the real price of oil into mutually orthogonal components with a clear structural economic interpretation. In practical terms, this approach relies on a structural vector autoregressive (SVAR) model based on monthly data for $z_t = (\Delta prod_t, rea_t, rpo_t)'$ where $\Delta prod_t$ denotes the percent change in global crude oil production, rea_t represents the index of real economic activity, and rpo_t refers to the real price of oil (see Appendix A). The structural VAR representation is given by

$$A_0 z_t = \alpha + \sum_{i=1}^{24} A_i z_{t-i} + \varepsilon_t, \quad (1)$$

where ε_t denotes the vector of serially and mutually orthogonal structural innovations, constructed following the decomposition of the reduced-form errors e_t according to $e_t = A_0^{-1} \varepsilon_t$. Under this decomposition, the reduced-form innovations can be written as

$$\begin{pmatrix} e_t^{\Delta prod} \\ e_t^{rea} \\ e_t^{rpo} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_t^{\text{oil supply}} \\ \varepsilon_t^{\text{aggregate demand}} \\ \varepsilon_t^{\text{oil-specific demand}} \end{pmatrix} \quad (2)$$

where the three structural shocks are respectively interpreted as oil supply shocks, aggregate demand shocks, and oil-specific demand shocks.

Importantly, the identification strategy is based on the following economic interpretations. First, a vertical short-run supply curve for crude oil is assumed, implying no contemporaneous response of production to changes in either aggregate demand or oil-specific demand. This is consistent with the economic intuition that oil-producing countries do not respond immediately, given the costs of adjusting oil production and the uncertainty surrounding the state of the crude oil market. Second, with respect to real economic activity, an exclusion restriction is imposed such that increases in the real price of oil driven by shocks that are specific to the oil market do not affect global real economic activity contemporaneously, but only with a delay of at least one month. Finally, the real price of oil is allowed to respond contemporaneously to all three orthogonal structural shocks, reflecting the high flexibility and forward-looking nature of oil prices in global markets.

2.2 Empirical Results

The reduced form of the VAR model is estimated using the least-squares method. The resulting estimates are then employed to construct the structural VAR representation of the model. Statistical inference is conducted using a recursive-design wild bootstrap with 2,000 replications.

Details on the data employed in the analysis are provided in Appendix A.

First, the replication results for the sample period from January 1974 to December 2007 are presented in Figures 1 and 2.

Figure 1 shows the responses of global oil production, real economic activity and the real price of oil to one-standard deviation structural innovations. All shocks have been normalized such that an innovation will tend to raise the price of oil. The results are consistent with those reported in

Kilian (2009). An unexpected oil supply disruption leads to an immediate and sharp decline in global oil production, which is only partially reversed within the first year, reflecting compensating production increases in other regions. In contrast, its effects on real economic activity and, more importantly, on the real price of oil are modest and, frequently, statistically insignificant. The aggregate demand shock generates an increase in oil production with a lag of about six months, while its effect on real economic activity is of the same sign, immediate, and highly persistent. Interestingly, the effect on the real price of oil is significant but not immediate, exhibiting a steadily increasing response over the 15-month horizon considered. Finally, unanticipated oil-market specific demand shocks exert an immediate, large, and persistent positive effect on the real price of oil, consistent with theoretical models of precautionary demand. These shocks are also associated with a temporary increase in real economic activity and a short-run decline in oil production, without generating any sustained increase in global oil production.

Interestingly, oil supply shocks do not generate a quantitatively relevant effect on the real price of oil. This finding is also confirmed in Figure 2, which reports the cumulative effects of each structural shock on the real price of oil. Consistently, the contribution of oil supply shocks to the real price of crude oil remains largely residual over time, without exhibiting any economically relevant impact. In contrast, aggregate demand shocks display a pronounced cyclical pattern. Finally, oil-market specific demand shocks account for the sharp increases and declines observed in the real price of oil, highlighting their central role in driving oil price dynamics and volatility.

It is worth mentioning that, although the results presented above are largely coherent with those reported in Kilian (2009), some differences emerge due to recent revisions in the real economic activity index, as proposed by Kilian (2019). In particular, the cumulative effect of the aggregate demand shock displays some discrepancies in terms of magnitude and trend, especially starting in the 1990s.

Going beyond the sample used in Kilian (2009) and extending the dataset time span until September 2025 allows for a robustness check of the model and empirical results presented in the original paper, as well as for highlighting new and potentially interesting findings. Figures 3 and 4 report the results for the extended version of the analysis.

Looking at Figure 3 a new and interesting piece of evidence arises. When considering the effect on the real price of oil, an unanticipated supply disruption generates, contrary to what is documented in Kilian (2009), a positive and significant effect over the entire horizon. Looking at aggregate demand shocks, two main elements emerge. First, the effect on real economic activity appears to be less persistent than in the original specification, but more intense at $t = 0$. Second, the impact of the shock on the real price of oil remains positive, but the increase occurs more rapidly, around $h = 3$, and then remains stable until $h = 15$. Finally, oil-market specific demand shocks display a particular effect on oil production, which adjusts in a positive and persistent manner. This result is of particular interest, as it suggests a response by producers to a positive shock in precautionary demand, which was not highlighted by Kilian (2009).

Figure 4 confirms this new evidence. The cumulative effect of an oil supply shock on the real price of crude oil appears more pronounced than in the baseline specification, although it remains the least important among the three shocks. This finding is consistent with the results discussed above, according to which oil supply shocks now generate an opposite, significant, and persistent effect on the real price of oil. Moreover, the cumulative response to an aggregate demand shock becomes substantially more volatile and less smooth over time, with a marked reduction in smoothness emerging after the Global Financial Crisis in 2008. Finally, the cumulative effect of oil-market specific demand shocks remains strongly volatile and considerably more pronounced than that associated with the other shocks, thereby confirming the insights originally suggested by Kilian (2009).

3 The Extended Model: Incorporating Inventories Expectations

Building on the milestone framework conceptualized by Kilian (2009), this section presents an extension of the original model discussed above. The analysis employs a newly constructed dataset containing expectations of crude oil inventories from the U.S. EIA, with the aim of better understanding how precautionary demand shocks affect the evolution of the real price of oil.

3.1 Introducing Inventory Expectations: A Theoretically Consistent Extension

Within the Kilian (2009) framework, the orthogonal innovation classified as *oil-specific demand shock* is of particular theoretical interest and empirical relevance. As discussed in Section 2, this shock is intended to capture shifts in precautionary demand, driven by uncertainty about future supply shortfalls relative to expected demand. Empirically, as confirmed by the replication results, this component emerges as the dominant and most volatile driver of real oil price fluctuations. However, in the original three-variable specification, this shock is effectively identified as a residual: it absorbs all price movements not explained by current supply or aggregate demand shocks. This inevitably raises concerns about whether the real price of oil fluctuations associated with the oil-specific demand shock can be coherently interpreted in light of the economic intuition proposed by Kilian (2009).

Considering the aforementioned economic interpretation, the oil-specific demand shock is characterized by two fundamental features. First, it is inherently forward-looking, as it reflects expectations about future market conditions rather than current realizations. Second, it operates at the intersection of supply and demand dynamics, representing anticipated imbalances between expected availability and expected consumption. Consequently, inventories, and specifically expectations about their future levels, constitute a natural candidate for capturing precautionary demand shifts driven by anticipated supply shortfalls. The underlying economic intuition is straightforward: a negative shock to expected inventories signals a tightening of future market conditions and rising concerns about the balance between supply and demand. Consistent with Kilian (2009), this anticipation of lower future availability prompts an increase in precautionary demand today, exerting upward pressure on the real price of oil. This constitutes the theoretical channel underpinning the present analysis.

Building on the intuition that expected inventories provide a more direct measure of shocks to the anticipated future equilibrium in the oil market, it has been developed an augmented version of the original SVAR model by incorporating an additional time series based on monthly expectations of U.S. crude oil inventories published by the EIA (see Appendix A).

Before turning to the theoretical structure of the newly developed structural vector autoregressive model, it is worth emphasizing some key features of the newly constructed expected crude oil inventories series that ensure internal consistency and offer clear econometric advantages. First, beyond being consistent with the forward-looking economic intuition underlying oil-specific demand shocks, focusing on expected rather than realized inventory levels helps decouple the inventory series from aggregate demand and, most importantly, from oil production. This mitigates simultaneity concerns, providing a cleaner recursive identification structure. Secondly, a structural feature renders the newly constructed time series particularly interesting. As discussed in Appendix A, the EIA releases expectations for end-of-period inventories of the current month during the first half of the month, and often even earlier. This implies that the reported expectation is only weakly affected by prices realized within the same month. As a result, this information structure supports the identification strategy adopted in the structural vector autoregressive (SVAR) framework.

Formally, the analysis relies on a four-variable specification based on monthly data:

$$z_t = (\Delta prod_t, rea_t, inv_t^{t+h}, rpo_t)', \quad (3)$$

where inv_t^{t+h} denotes the expectation, formed at time t , of the logarithm of U.S. crude oil inventories at horizon $t + h$. The structural VAR representation is given by

$$A_0 z_t = \alpha + \sum_{i=1}^{24} A_i z_{t-i} + \eta_t, \quad (4)$$

where η_t is the vector of serially uncorrelated and mutually orthogonal structural innovations. These shocks are obtained from the reduced-form errors u_t via the relation $u_t = A_0^{-1} \eta_t$. Under this decomposition, the reduced-form innovations can be written as

$$\begin{pmatrix} u_t^{\Delta prod} \\ u_t^{rea} \\ u_t^{inv} \\ u_t^{rpo} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix} \begin{pmatrix} \eta_t^{\text{oil supply}} \\ \eta_t^{\text{aggregate demand}} \\ \eta_t^{\text{oil-specific precautionary demand}} \\ \eta_t^{\text{oil-specific residual demand}} \end{pmatrix}. \quad (5)$$

Beyond the standard restrictions established by Kilian (2009), the specific identification of the reduced-form innovations presented here is grounded on both economic intuition and structural characteristics of the data, namely, the timing of information release. Production is assumed not to respond contemporaneously to the oil-specific precautionary demand shock. Following the intuition in Kilian (2009), adjustments in crude oil production by producers are considered sluggish, reflecting a vertical short-run supply curve. Real economic activity is also assumed not to react immediately to oil-specific precautionary demand shocks. This is a non-trivial restriction, as it rests on the premise that real economic activity is primarily driven by realized economic conditions rather than by expectations about future shortfalls or surpluses in crude oil inventories. Expected inventories are allowed to respond contemporaneously to oil supply shocks and aggregate demand shocks. At first glance, this assumption may appear at odds with the timing of the release of expectations. However, a strong identifying assumption is required, particularly for aggregate demand shocks. The underlying idea is that, when forming expectations at time t for inventories at horizon $t + h$, the EIA already observes very short-run developments, on the order of a few weeks, in both production and real economic activity. While this assumption is arguably reasonable for the production side, given its relatively predictable nature, it may be more controversial when applied to aggregate demand shocks. Finally, real oil price fluctuations are assumed to respond immediately to changes in inventory expectations, consistent with the economic interpretation of the inventory-expectation shock and its timing. By contrast, expected inventories are not allowed to respond contemporaneously to the residual oil-market specific demand shock, in line with the timing of the data release.

Finally, before proceeding to the estimation of the new specification, it is necessary to clarify the choice of the lead h associated with inventory expectations. As described in Appendix A, the constructed dataset provides at least 13 forecast horizons for each month. Consistent with the economic interpretation of inventory expectations as a forward-looking variable, very short horizons close to $h = 0$ are not appropriate, as they would be more likely to reflect contemporaneous market conditions. At the same time, very distant horizons risk diluting relevant information and weakening the link with current market dynamics. On these grounds, forecast horizons between 4 and 8 months ahead appear economically reasonable. For clarity and simplicity, the analysis focuses on $h = 6$. This choice follows experimentation with alternative horizons within this range, which yielded no qualitatively different or statistically significant results.

3.2 Empirical Results

Consistent with Kilian (2009), the four-variable specification is estimated as follows. The reduced-form VAR is consistently estimated by least squares, and the resulting estimates are used to construct the structural VAR representation of the model. Statistical inference is conducted using a recursive-design wild bootstrap with 2,000 replications.

Figures 7 and 8 present the main findings of the extended model estimated over the sample period 2006.1–2025.9.¹

Figure 7 reports the impulse response functions following a one-standard-deviation structural shock. All shocks are normalized so as to generate a positive impact effect on the real price of oil. Two main insights emerge from this figure. First, the impulse response functions already discussed are coherent with the results obtained from the original three-variable benchmark. This close resemblance provides a positive robustness check for the identification strategy adopted in the original framework proposed by Kilian (2009). Second, the impulse responses involving expected inventories are fully consistent with economic intuition. In particular, oil production displays a positive and persistent response to a negative inventory-expectation shock. This result is coherent with the idea that oil producers coordinate their production decisions to smooth temporary market imbalances and to respond to expected changes in market conditions. Real economic activity, as expected, exhibits a small positive and mitigated response after impact, with no pronounced or persistent deviation from trend. By contrast, an oil-specific precautionary demand shock generates an immediate, strong, and persistent increase in expected inventories, reflecting heightened concerns about future market conditions. Finally, an inventory-expectation shock has a positive, statistically significant, and persistent effect on the real price of oil, fully consistent with the underlying economic intuition. Even when considering how the other structural shocks affect inventory expectations, the economic interpretation remains coherent. A negative oil supply shock leads to a persistent decline in expected inventories, consistent with tighter current and expected future supply conditions. A positive aggregate demand shock also generates a negative and persistent response of inventory expectations, reflecting the anticipated absorption of inventories due to stronger global demand. Finally, the residual shock similarly induces a decrease in expected inventories, reinforcing the interpretation of this component as capturing precautionary or market-specific forces affecting expectations.

The historical decomposition of the real price of crude oil presented in Figure 8 largely confirms the structural decomposition obtained from the original three-variable model, thereby supporting the robustness of the identified oil supply and aggregate demand shocks. Additionally, the newly introduced oil-specific precautionary demand shock evolves smoothly over most of the sample period, with a pronounced increase in its contribution around the COVID-19 pandemic.

Consistent with the theoretically grounded approach adopted in this paper, the inclusion of the expected inventories time series and the identification of a new oil-specific precautionary demand shock should be interpreted in direct relation to both the historical contribution of the fourth shock in the extended model – the so-called oil-specific residual shock – and the third shock in the framework of Kilian (2009) – labelled the oil-specific demand shock. Focusing on the historical contributions – more clearly illustrated in Figure 9 – two main patterns emerge. First, the historical contribution of the newly identified residual shock appears noticeably less volatile than the oil-specific demand shock in the original three-variable SVAR. Second, the very large contribution of the oil-specific demand shock identified in the Kilian (2009) framework during episodes of heightened uncertainty is substantially attenuated once the new residual shock is considered. Taken together, these results suggest that the newly introduced time series, and the associated

¹To establish a consistent benchmark, it has been also replicated the estimation of the original three-variable model from Kilian (2009) over the same sample period 2006.1–2025.9, with the results reported in Figures 5 and 6.

oil-specific precautionary demand shock, plays an important role in partially explaining dynamics that were previously captured by the oil-specific demand shock in the original framework. This role is particularly relevant during periods characterized by extreme volatility and heightened uncertainty, further strengthening the coherence with the intuition presented by Kilian (2009).

Beyond the graphical representation provided in Figure 9, several quantitative metrics are particularly informative for assessing whether, and to what extent, the newly developed oil-specific precautionary demand shock is able to partially describe the contribution and volatility of the oil-specific demand shock originally identified by Kilian (2009). Specifically, these measures allow a more formal evaluation of how much of the magnitude, persistence, and variability previously attributed to the oil-specific demand shock is absorbed by the new channel introduced in the extended model.

Firstly, the variance reduction associated with the cumulative effect has been computed by comparing the variance of the oil-specific residual shock from the extended model with the variance of the oil-specific demand shock identified in the original framework of Kilian (2009). The variance is defined as

$$\text{Var}(H^{rpo,j}) = \frac{1}{T-1} \sum_{t=1}^T \left(H_t^{rpo,j} - \overline{H}^{rpo,j} \right)^2, \quad j \in \{3, 4\}. \quad (6)$$

where $j = 3$ refers to the oil-specific demand shock in the three-variable model, while $j = 4$ denotes the oil-specific residual shock in the four-variable specification. Based on this measure, the variance reduction is computed as

$$\text{Variance Reduction} = 1 - \frac{\text{Var}(H^{rpo,4})}{\text{Var}(H^{rpo,3})}. \quad (7)$$

The results are strongly interesting, showing a variance reduction of 32.34%. This outcome confirms what was already graphically visible: the newly identified oil-specific residual shock is markedly less volatile than the oil-specific demand shock proposed by Kilian (2009). This evidence suggests that the new specification is able to better account for the cumulative effect volatility previously attributed to the oil-specific demand shock, capturing dynamics related to production, aggregate demand, and, most importantly, oil-specific precautionary demand.

Once the volatility issue has been addressed, another element of interest concerns the level of contribution estimated for the oil-specific residual shock relative to the oil-specific demand shock identified by Kilian (2009). To evaluate this aspect, the following measure is constructed, which simply considers the absolute value of the cumulative effect at each point in time and sums it over the sample period. More specifically,

$$\mathcal{A}^{rpo,j} = \sum_{t=1}^T \left| H_t^{rpo,j} \right|, \quad j \in \{3, 4\}. \quad (8)$$

where $j = 3$ refers to the oil-specific demand shock in the original three-variable model and $j = 4$ denotes the oil-specific residual shock in the extended specification. Once this index is computed, a ratio is constructed as

$$\text{Area Ratio} = \frac{\mathcal{A}^{(4)}}{\mathcal{A}^{(3)}}. \quad (9)$$

Interestingly, the estimated ratio is equal to 0.929, suggesting that, even though the cumulative effect of the residual shock is now less volatile, it still plays a substantial role in the determination of the real price of crude oil, even within the new model specification.

Finally, a last metric has been constructed with the aim of understanding whether the newly identified oil-specific precautionary demand shock actively contributes to partially explaining the oil-specific demand shock originally identified by Kilian (2009). To approximate this idea, a simple

metric is employed, which considers the difference in cumulative contribution between the two residuals and then evaluates its correlation with the inventory-based shock. Formally, the residual gap is defined as

$$\Delta H_t^{rpo} = H_t^{rpo,3} - H_t^{rpo,4}. \quad (10)$$

where $H_t^{rpo,3}$ denotes the oil-specific demand shock from the three-variable model, and $H_t^{rpo,4}$ denotes the oil-specific residual shock from the extended specification. The relationship between this residual gap and the inventory shock is then assessed through the following correlation measure:

$$\rho_{\Delta H^{rpo}, H^{rpo,inv}} = \text{Corr}(\Delta H_t^{rpo}, H_t^{rpo,inv}). \quad (11)$$

where $H_t^{rpo,inv}$ denotes the cumulative contribution of oil-specific precautionary demand shock. The result is extremely interesting: the estimated correlation is equal to 0.8012, indicating that the difference between the contribution from the original oil-specific demand shock and the new residual shock is strongly driven by the oil-specific precautionary demand component. This evidence supports the interpretation that the new specification actively captures a substantial portion of the dynamics previously absorbed by the oil-specific demand shock.

4 Conclusions

The current paper extends the framework proposed by Kilian (2009), with the objective of improving the identification of oil price drivers by capturing the precautionary demand component that, in Kilian’s original framework, is subsumed within the residual oil-specific demand shock. To this end, a time series of expected crude oil inventories at horizon $h = 6$ is utilized as a proxy for forward-looking expectations about equilibrium conditions in the oil market.

In addition to providing a strong robustness check of the original estimation—through replication and by extending the sample up to September 2025—the analysis demonstrates that identifying an oil-specific precautionary demand shock via inventory expectations sheds new light on the cumulative drivers of the real price of oil. Furthermore, it reveals a high correlation between the cumulative effect of the newly identified precautionary demand shock and the gap between the cumulative contribution of the original residual shock and that of the new residual shock. This suggests that the new shock is not capturing extraneous noise, but rather giving structural form to the speculative dynamics that drove the residual in the benchmark model.

Taken together, these findings suggest that explicitly accounting for expectations about future inventory conditions significantly improves the structural interpretation of oil price dynamics. The extended model successfully disentangles genuine precautionary demand from residual noise, proving that what appeared as an unexplained residual in the original framework is partially driven by identifiable shifts in inventory expectations.

Appendix

A Data

Most of the data used in this paper are closely related to those employed in Kilian (2009). However, some necessary clarifications and additional descriptions are provided below.

Two key deviations from the original Kilian (2009) specification were adopted to ensure consistency and comparability across the different specifications and sample selections. First, due to data availability constraints, the estimation window begins in January 1974 rather than February 1973. Second, while the definitions of crude oil production and real oil prices remain conceptually identical, we employ the log-levels of the raw series without the specific historical transformations originally used by Kilian for graphical purposes, opting instead for a uniform measurement approach. Additionally, the index of real economic activity (REA) has been updated following the revisions proposed by Kilian (2019) to better reflect recent global trade dynamics.

What follows is a detailed description of each time series used in the analysis.

Crude Oil Production Global crude oil production is defined as world crude oil production including lease condensate, measured in thousand barrels per day. The data are sourced from the U.S. Energy Information Administration (EIA) at a monthly frequency. In the econometric analysis, this variable enters the model in log-differences to capture the growth rate of global oil supply.

Real Economic Activity The measure of real economic activity is based on the index originally introduced by Kilian (2009), which is designed to capture cyclical fluctuations in the global demand for industrial commodities. This index is constructed from a panel of dry cargo single-voyage ocean freight rates for bulk commodities (such as grain, coal, and iron ore). The raw nominal rates are deflated by the U.S. CPI and then linearly detrended to remove long-run technological trends in the shipping industry, thereby isolating the cyclical component of demand. In this paper, an updated version of this index is employed. More specifically, the analysis uses the revised series proposed by Kilian (2019) and provided by the Federal Reserve Bank of Dallas.

Real Price of Crude Oil The real price of crude oil is constructed using the refiner acquisition cost of imported crude oil as reported by the U.S. Energy Information Administration (EIA). This nominal price series is deflated using the U.S. Consumer Price Index (CPI, 2008 = 100) obtained from Federal Reserve Economic Data (FRED). The variable enters the model in log-levels.

Expected Inventories The expected inventories dataset has been constructed using the archive of the Short-Term Energy Outlook (STEO) published by the U.S. Energy Information Administration (EIA). Each monthly release provides an Excel spreadsheet reporting both historical data and expectations for a wide range of variables related to energy markets. In this paper, the series corresponding to U.S. crude oil inventories (excluding the Strategic Petroleum Reserve) has been selected.²

The final dataset constructed from the STEO archive provides, for each month, at least 13 forecasts corresponding to horizons ranging from the current month up to 13 months ahead and spans the period from 2006.1 to 2025.9. It is worth noting that the expectation at horizon $h = 1$ refers to the current month, since the STEO release date always occurs before the middle of the month and often

²it has been chosen to focus on U.S. crude oil inventories rather than OECD inventories for three main reasons: (a) crude oil inventories are consistently available only at the U.S. level; (b) the U.S. series is longer, extending back to 2006; (c) given that the real price of oil used in the analysis is based on U.S. data, focusing on the U.S. level should not introduce major distortions.

before the tenth day. As a result, these forecasts can be interpreted as genuinely forward-looking expectations formed within the same month.

B Inverting the variables order: a robustness check

In the baseline specification, the variables are ordered as $\Delta prod$, rea , inv , and rpo . This ordering implicitly assumes that the real price of oil responds contemporaneously to oil-specific precautionary demand shocks, while inventories do not respond immediately to oil-specific residual demand shocks. As a robustness check, the ordering of the last two variables is inverted, placing the real price of oil before expected inventories. The results of this robustness exercise are reported in Figures 12 and 13.

The impulse response functions shown in Figure 12 indicate that the qualitative patterns remain very similar to those obtained in the baseline specification, with one noteworthy difference. Under the inverted ordering, the real price of oil is no longer significantly affected by the precautionary demand shock, whereas expected inventories respond immediately and persistently to shocks associated with oil-specific residual demand. This change in contemporaneous responses is consistent with the alternative timing assumptions embedded in the inverted ordering.

Importantly, the robustness check is successful. The historical decomposition presented in Figure 13 closely mirrors the patterns observed in the baseline results. The relative contributions of the different shocks remain stable over time, confirming that the disentanglement between oil-specific residual demand and precautionary demand shocks is robust to changes in the ordering of the variables. Overall, this evidence supports the coherence and consistency of the proposed identification strategy.

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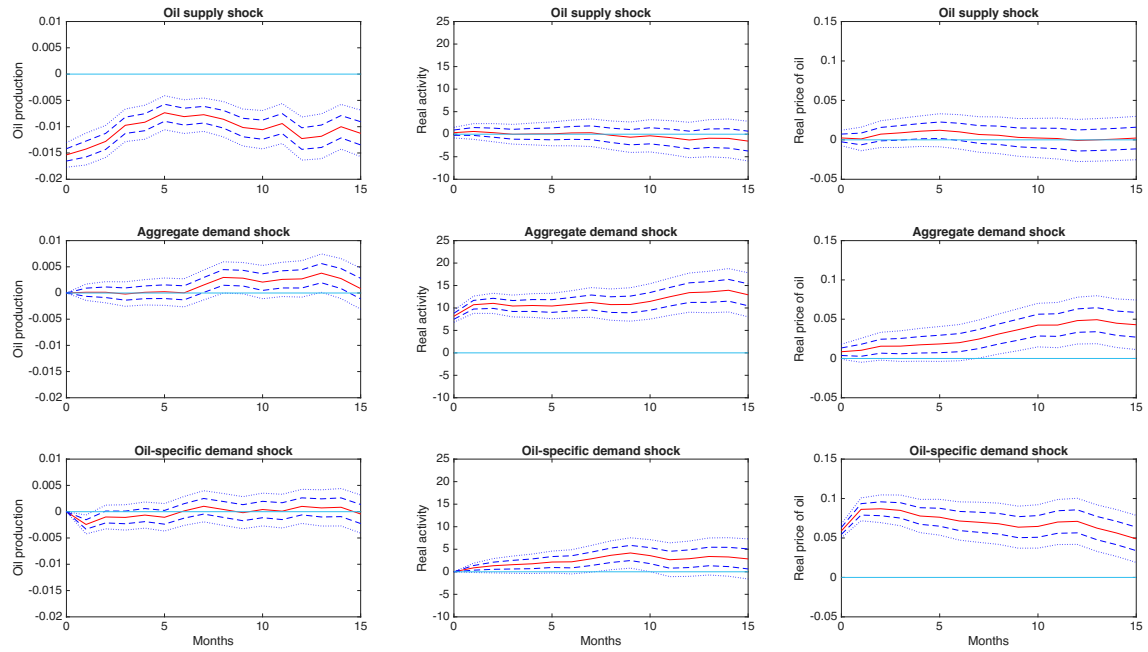


Figure 1: Responses to One-Standard Deviation (1974.1-2007.12)

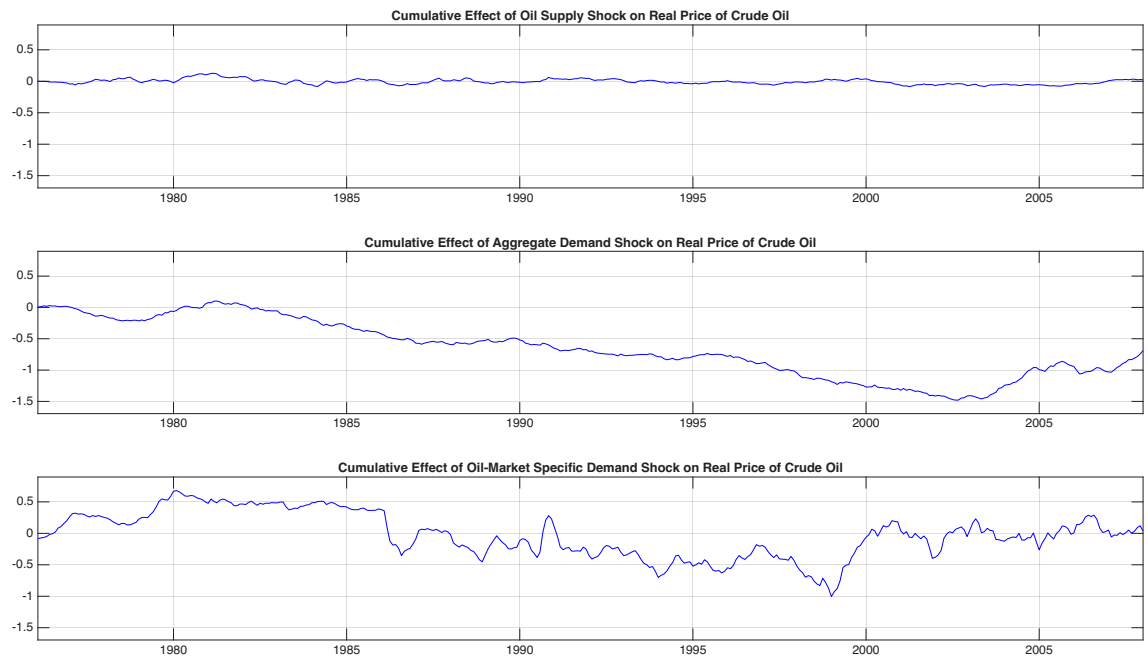


Figure 2: Historical Decomposition of Real Price of Oil (1976.1-2007.12)

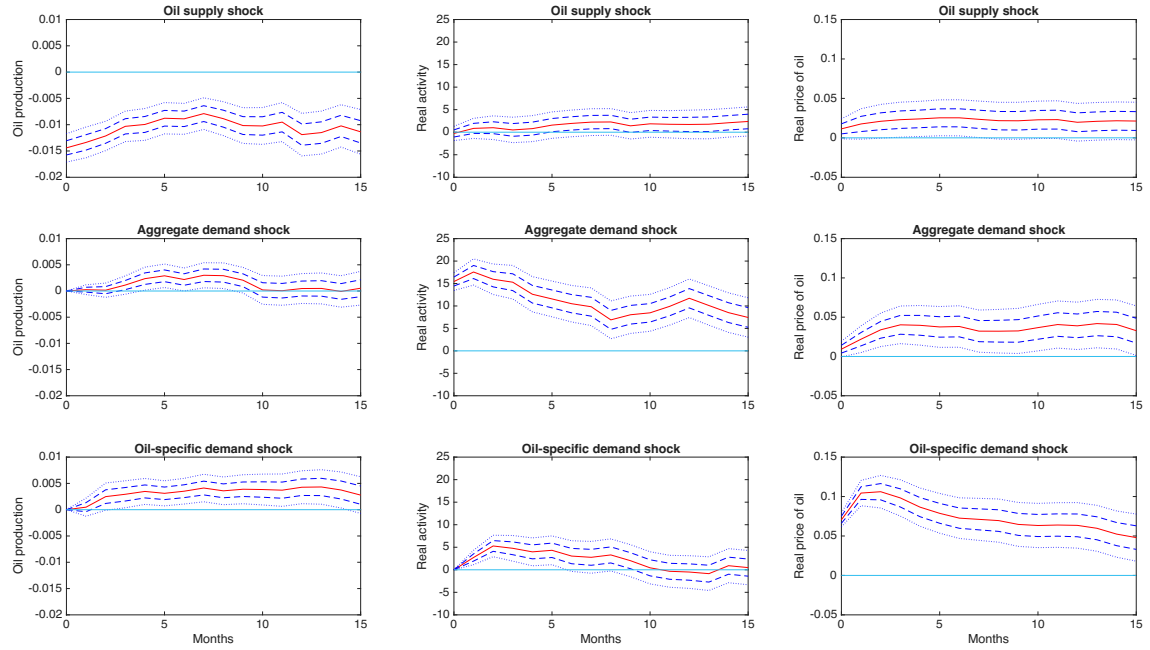


Figure 3: Response to One-Standard Deviation (1974.1-2025.9)

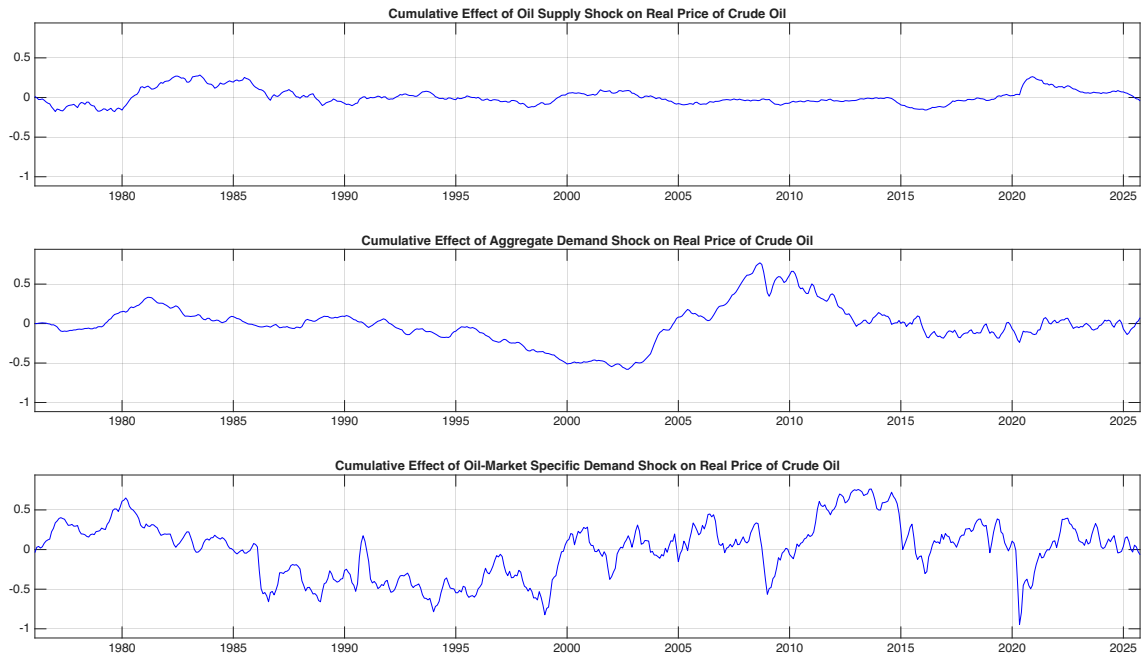


Figure 4: Historical Decomposition of Real Price of Oil (1976.1-2025.9)

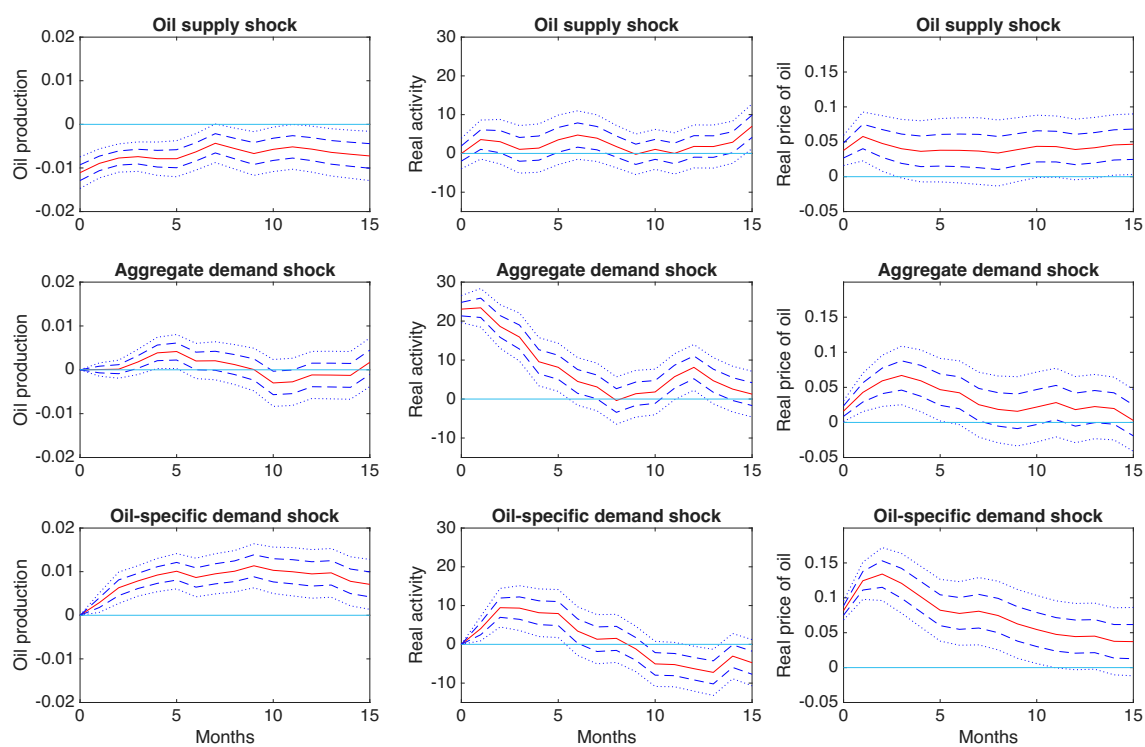


Figure 5: Response to One-Standard Deviation (2006.1-2025.9)

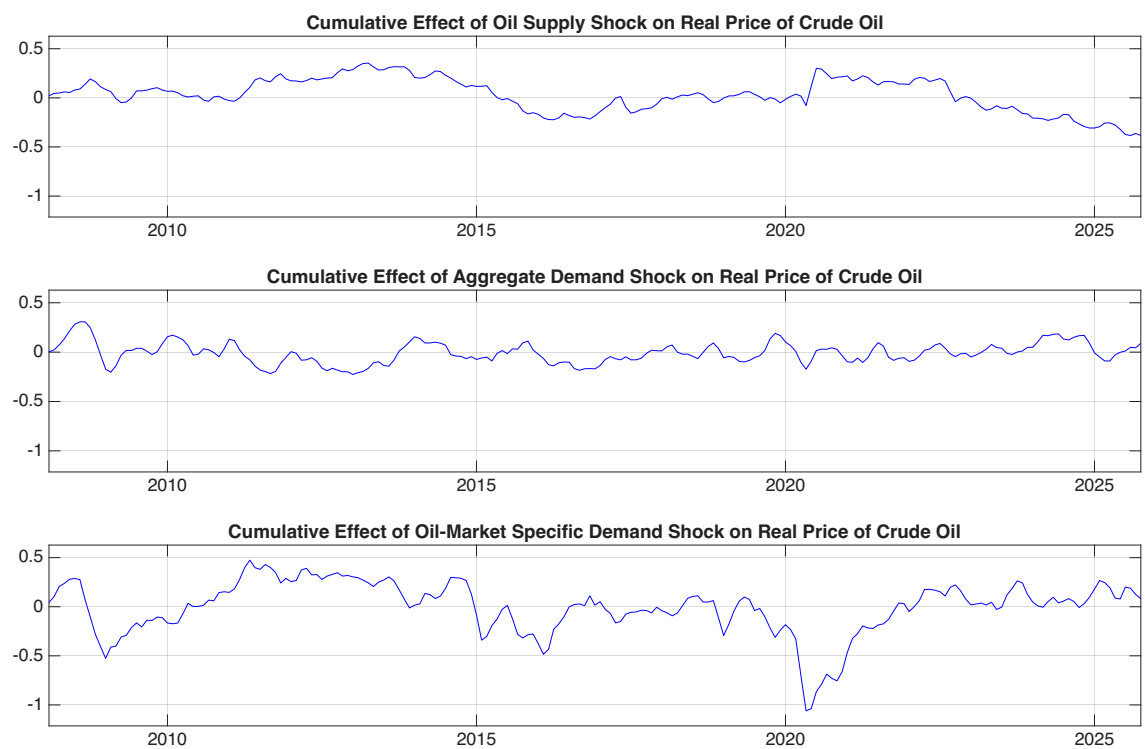


Figure 6: Historical Decomposition of Real Price of Oil (2006.1-2025.9)

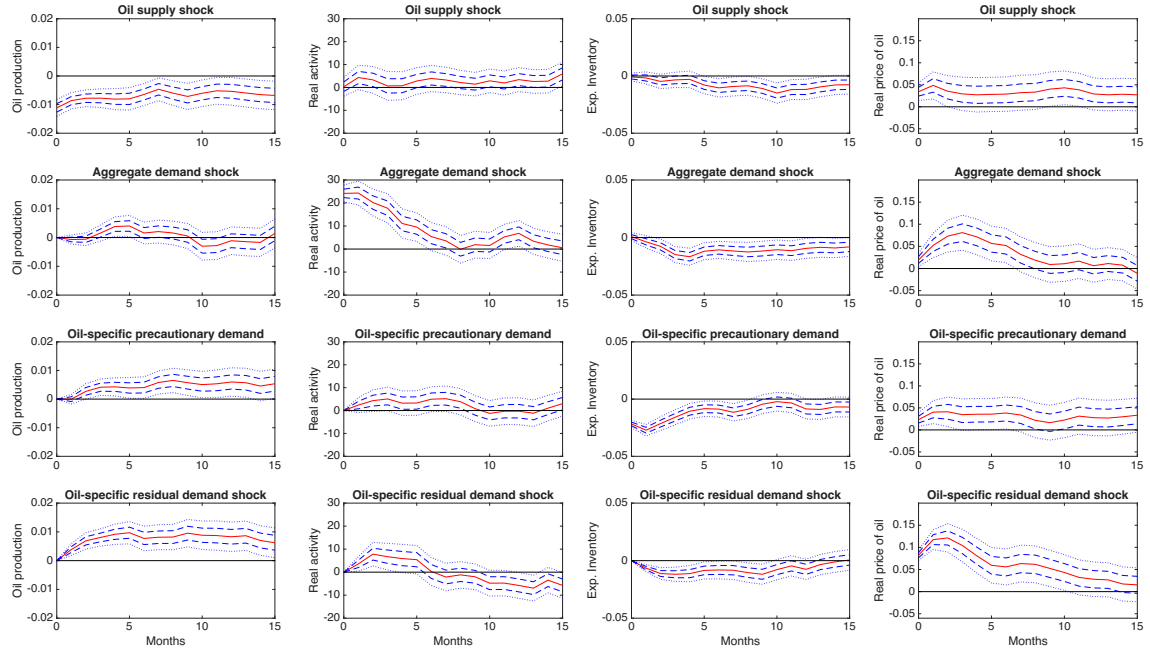


Figure 7: Response to One-Standard Deviation (2006.1-2025.9)

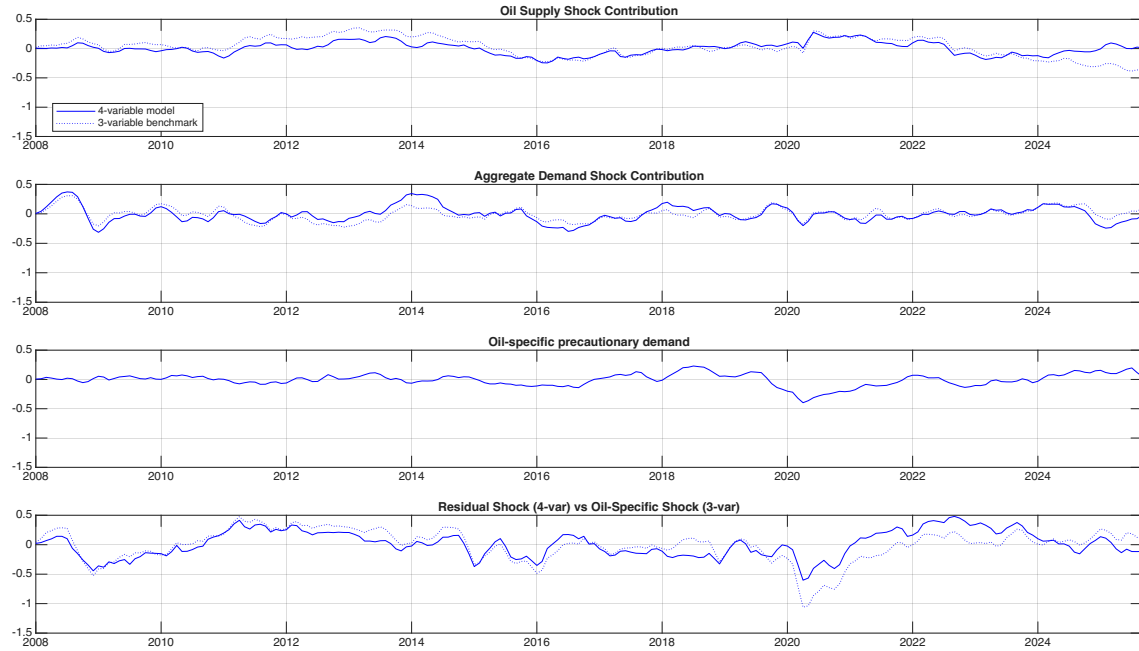


Figure 8: Historical Decomposition of Real Price of Oil (2006.1-2025.9)

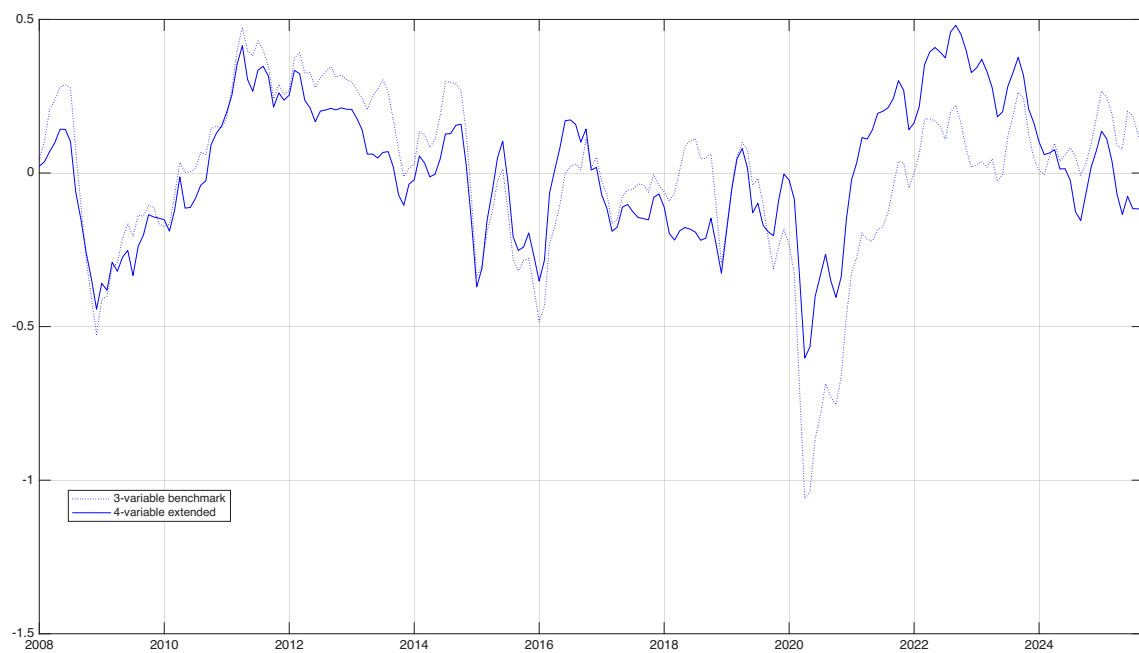


Figure 9: Cumulative Effect of Residual Shock on Real Price of Oil (2006.1-2025.9)

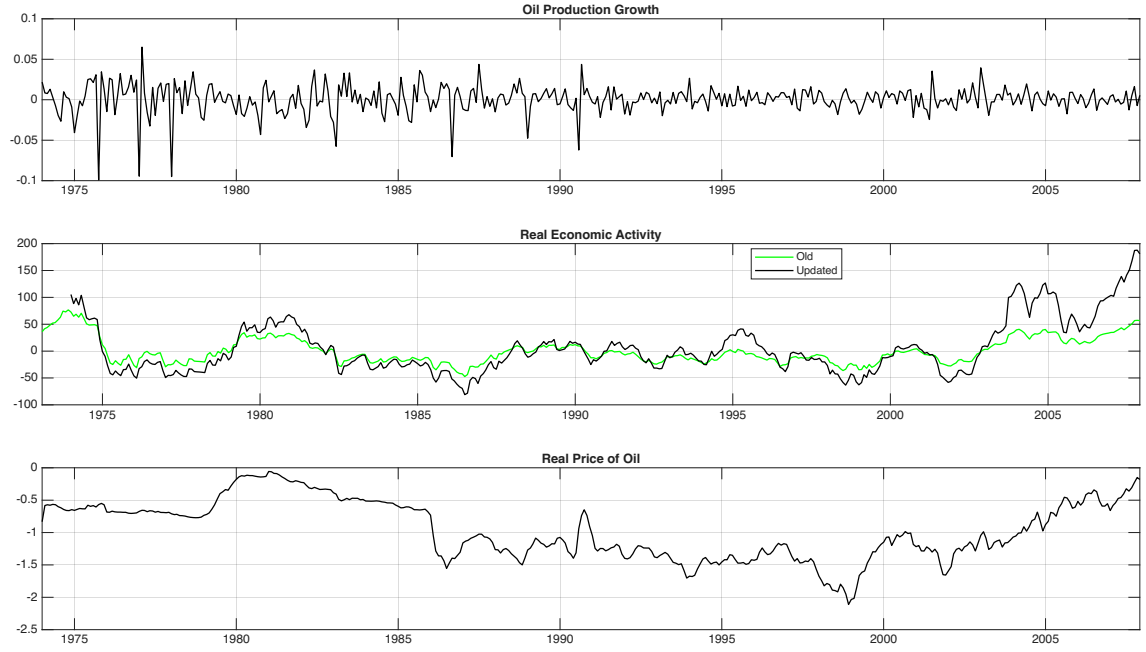


Figure 10: Baseline transformed data: $\Delta prod$, rea , and rpo (1974.1–2007.1)

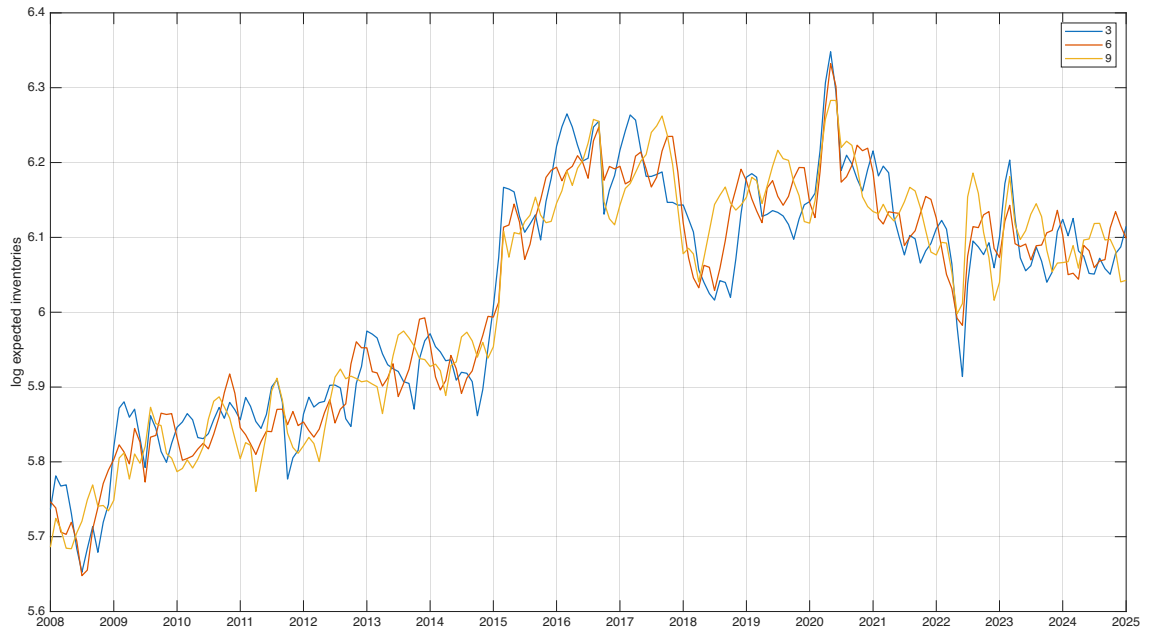


Figure 11: Expected inventories at $h = 3, 6, 9$ (2008.1–2025.9)

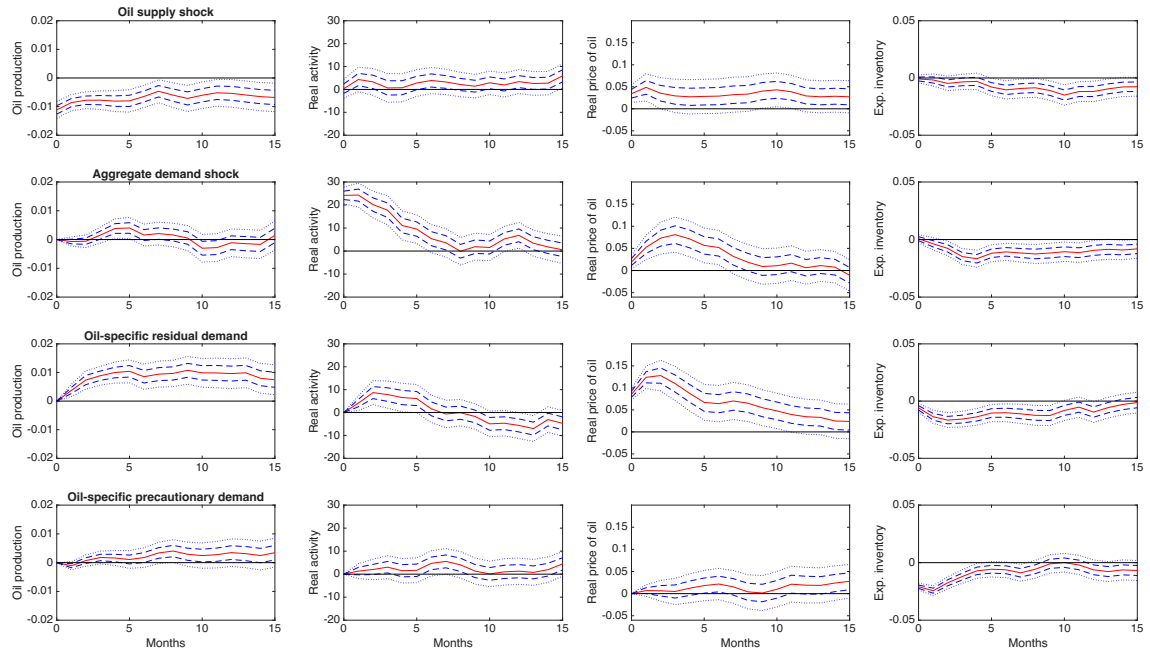


Figure 12: Response to One-Standard Deviation (2006.1-2025.9)

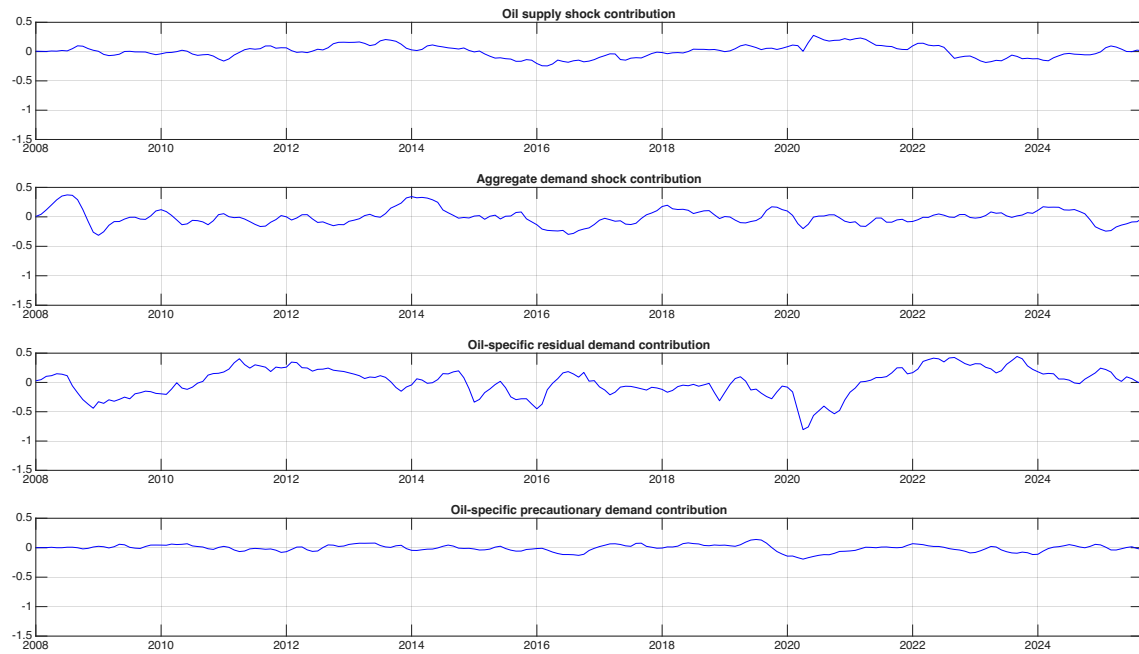


Figure 13: Historical Decomposition of Real Price of Oil (2006.1-2025.9)